

KOPPEN'S SCHEME OF CLASSIFICATION OF CLIMATE

[illegible]

Code	Humidity Category	Basis
A	Humid	Temperature-based
B	Dry	Precipitation-based
C	Humid	Temperature-based
D	Humid	Temperature-based
E	Humid	Temperature-based

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Secondary Code

Letter	Meaning
f	No dry season
m	Monsoon climate
w	Winter dry season
s	Summer dry season
S	Steppe or semi-arid
W	Desert

CLIMATIC TYPES ACCORDING TO KOPPEN

	Climate Type	Code	Latitude	Key Feature
1	Tropical Wet (Equatorial)	Af	5°N-5°S	Hot, wet year-round; no dry season
2	Tropical Monsoon	Am	5°-25°	Heavy summer rain; short dry winter; seasonal reversal of winds
3	Tropical Savanna	Aw	5°-20°	Wet summer, dry winter; grasslands with scattered trees
4	Subtropical Steppe	BSh	15°-35°	Hot semi-arid; low unreliable rainfall; desert margin
5	Subtropical Desert	BWh	15°-30°	Hot desert; <250 mm rain; extreme temperatures
6	Mid-Latitude Steppe	BSk	35°-60°	Cold semi-arid; warm summers, cold winters; continental interior
7	Mid-Latitude Desert	BWk	35°-60°	Cold desert; very cold winters; strong continentality
8	Humid Subtropical (China Type)	Cfa	25°-40° eastern margins	Hot humid summers; mild winters; year-round rain
9	Mediterranean	Cs	30°-45° western margins	Hot dry summer; mild wet winter
10	Marine West Coast (British Type)	Cfb	40°-60° western margins	Cool summers, mild winters; year-round rain; oceanic
11	Humid Continental (Laurentian Type)	Df	40°-60° eastern/interior	Warm summers, cold winters; large annual range
12	Subarctic (Taiga/Boreal)	Dw	50°-70°N	Long severe winters; short summers; coniferous forests; podzol
13	Tundra	ET	Arctic margins	Very cold; warmest month 0°-10°C; permafrost; no trees

14	Polar Ice Cap	EF	Antarctica, Greenland	Below 0°C year-round; polar desert; permanent ice
15	Highland	H	High altitudes, any latitude	Altitude-controlled; altitudinal zonation of vegetation

1. EQUATORIAL CLIMATE

Tropical Wet Climate - Köppen Code: Af

Distribution

Found mainly between **5°N and 5°S**, extending to about 10° in some areas. Main regions: **Amazon Basin, Congo Basin, Indonesia, Malaysia and New Guinea.**

Cause of Formation

The Equator receives intense solar heating throughout the year because the Sun remains high in the sky. The equatorial low-pressure belt and ITCZ remain close to this region. Trade winds converge and warm, moisture-laden air rises continuously, cools, condenses and produces heavy convectional rainfall.

Temperature and Rainfall

Parameter	Detail
Mean Temperature	25-27°C throughout the year. No true winter.
Annual Temperature Range	Very small - about 2-3°C.
Daily Range	Often greater than the annual range.
Annual Rainfall	Generally above 2,000 mm. No distinct dry season.
Relative Humidity	Generally above 80%.
Pressure and Winds	The equatorial low-pressure belt dominates. Trade winds converge near the ITCZ. Surface winds are generally weak; strong vertical movement.

Vegetation, Soil and Agriculture

Natural vegetation is **tropical evergreen rainforest** with several layers. Trees are tall and straight with buttress roots, drip-tip leaves, lianas and epiphytes. Heavy rainfall causes **intense leaching**; rapid decomposition of organic matter means soil nutrients are quickly absorbed by vegetation - the soil itself is often relatively poor in nutrients. Traditional shifting cultivation occurs in some areas. Commercial crops: **rubber, cocoa, oil palm and banana.**

Important Features

Hot, wet and humid throughout the year. Very little seasonal temperature variation. No distinct dry season. Heavy convectional rainfall. Dense evergreen rainforest.

2. TROPICAL MONSOON CLIMATE

Köppen Code: Am

Distribution

Found mainly in **South and Southeast Asia**: India, Bangladesh, Myanmar, Thailand, Vietnam, southern China and parts of northern Australia. Smaller areas in tropical Africa and tropical America.

Cause of Formation

Develops due to the **seasonal reversal of pressure and winds**. In summer, land heats faster than the ocean, develops low pressure and draws moisture-laden winds from the ocean (ITCZ shifts landward). In winter, land cools faster, develops high pressure and winds blow from land to sea. Mountains strongly modify rainfall.

Temperature and Rainfall

Parameter	Detail
Mean Annual Temperature	Generally above 20°C. Hot humid summers; relatively cooler winters.
Annual Rainfall	Generally 1,000-2,500 mm. Most rainfall in summer; short dry winter season.
Nature of Rainfall	Mainly orographic (where winds cross mountains). Also convectional and tropical cyclone-associated.

Vegetation, Soil and Agriculture

Natural vegetation is **tropical deciduous or monsoon forest**. Important trees: teak, sal, bamboo and sandalwood. Heavy rainfall causes leaching in wetter areas; alluvial soils are fertile in river plains. Agriculture is intensive: **rice** in wetter areas; tea, jute, sugarcane and cotton are also important.

Important Features

High temperature, seasonal reversal of winds, heavy summer rainfall and a short dry season. Rainfall is strongly seasonal compared to equatorial climate.

3. TROPICAL WET AND DRY CLIMATE

Tropical Savanna Climate - Köppen Code: Aw

Distribution

Occurs mainly between about **5° and 20° north and south of the Equator**. Lies between the equatorial wet climate and tropical dry regions. Most extensive in Africa. Major regions: **Sudan and East Africa, Llanos of Venezuela and Colombia, Campos of Brazil and northern Australia**.

Cause of Formation

Develops mainly due to the **seasonal movement of the ITCZ**. During summer, ITCZ shifts over the region and brings moist, rising air producing convectional rainfall. During winter, ITCZ moves away and the region comes under dry trade winds and subtropical high pressure.

Temperature and Rainfall

Parameter	Detail
Temperature	High throughout the year. Mean monthly temperatures generally above 20°C.
Annual Rainfall	Generally 500-1,500 mm. Most rainfall in summer; winter is distinctly dry.
Seasons	Two distinct seasons: hot wet summer and warm dry winter. The wet season is longer near the equator.

Vegetation, Wildlife, Soil and Agriculture

Natural vegetation is **tropical grassland with scattered trees**. Trees: acacia and baobab. Wildlife: zebra, giraffe, antelope, elephant, buffalo, lions, leopards and cheetahs. Soils are generally more fertile than heavily leached equatorial soils. Crops: millets, sorghum, maize, groundnut and cotton.

Important Features

High temperature throughout the year. Distinct wet summer and dry winter. Extensive grasslands with scattered trees. Seasonal movement of ITCZ is the main control of rainfall.

4. SUBTROPICAL STEPPE CLIMATE

Hot Semi-Arid Climate - Köppen Code: BSh

Distribution

Occurs mainly on the **margins of tropical and subtropical deserts**. Major regions: Sahel of Africa, margins of the Sahara and Arabian deserts, parts of India, interior Australia, southern Africa and parts of Mexico.

Cause, Temperature and Rainfall

Develops in the transition zone between humid tropical regions and hot deserts. The subtropical high-pressure belt produces dry descending air for much of the year. Seasonal movement of the ITCZ brings limited rainfall during warmer months. Temperature remains high; summers very hot, winters warm or mild. Annual rainfall: **250-500 mm**, low, irregular and unreliable.

Vegetation and Key Features

Short grasses, thorny bushes and scattered drought-resistant trees. Overgrazing and removal of vegetation can lead to **desertification**. Forms a transition between tropical grasslands and hot deserts.

5. SUBTROPICAL DESERT CLIMATE

Hot Desert Climate - Köppen Code: BWh

Distribution

Occurs mainly between **15° and 30° north and south**, especially on western margins and interiors of continents. Major regions: **Sahara, Arabian, Thar, Namib, Atacama and Australian Deserts**.

Cause, Temperature and Rainfall

Most subtropical deserts lie under the **subtropical high-pressure belt**. Descending air around 20°-30° latitude becomes warm and dry, preventing cloud formation. Cold ocean

currents increase aridity along western continental margins. Summers are extremely hot - daytime temperatures frequently exceed **40°C**. Large daily temperature range. Annual rainfall generally **below 250 mm**; highly irregular.

Vegetation and Key Features

Very sparse xerophytic vegetation. Plants develop deep roots; leaves may be reduced to spines; succulents store water. Agriculture concentrated around oases and irrigated areas. **Extremely low rainfall, high temperature, large daily temperature range, clear skies, low humidity** and sparse vegetation characterise this climate.

6. MID-LATITUDE STEPPE CLIMATE

Cold Semi-Arid Climate - Köppen Code: BSk

Distribution

Occurs mainly in the **interiors of continents in the middle latitudes**: Central Asia, western North America, Patagonia and parts of eastern Europe.

Cause, Temperature and Rainfall

Develops mainly due to **continentality** - far from oceans, so moisture-bearing winds lose much moisture before reaching them. Rain-shadow effects behind major mountain ranges also increase dryness. Summers warm to hot; winters cold to very cold. Annual rainfall: **250-500 mm**.

Vegetation and Key Features

Short temperate grasses with extensive root systems. Many steppe soils are **fertile** and suitable for cultivation. Wheat and other cereals are important; commercial grain farming and livestock rearing are widespread. Differs from subtropical steppe mainly because of **colder winters** and stronger continental influence.

7. MID-LATITUDE DESERT CLIMATE

Cold Desert Climate - Köppen Code: BWk

Distribution

Occurs mainly in the **interiors of continents in the middle latitudes**: Gobi Desert, Taklamakan Desert, Great Basin of North America, parts of Patagonia and Central Asia.

Cause, Temperature and Rainfall

Develops mainly due to **continentality and rain-shadow effects**. Summers warm to hot; winters very cold with temperatures falling below freezing. The annual range is very large. Annual rainfall **below 250 mm**; some precipitation as snow in winter. Differs from hot deserts mainly because of **severe winter cold**.

8. HUMID SUBTROPICAL CLIMATE

Warm Temperate Eastern Margin / China Type - Köppen Code: Cfa

Distribution

Occurs mainly between **25° and 40° latitudes on the eastern margins of continents**. Major regions: south-eastern China, southern Japan, south-eastern USA, eastern Australia, south-eastern Brazil, Uruguay and northern Argentina.

Cause, Temperature and Rainfall

Develops on eastern continental margins where warm ocean currents and moist onshore winds provide abundant moisture. Strong summer heating produces convection and heavy

rainfall; tropical cyclones also bring heavy rain in some regions. Summers are hot and humid (25-30°C); winters mild to cool. Annual rainfall: **750-1,500 mm**, with summer maximum; no pronounced dry season.

Vegetation, Soil and Agriculture

Subtropical evergreen and temperate deciduous forests. Important trees: oak, magnolia, camphor, bamboo. Long growing season and abundant rainfall favour intensive cultivation: **rice, tea, cotton, sugarcane and fruits**. Dense populations, intensive agriculture and major industrial centres characterise these regions.

Important Features

Hot humid summers, mild to cool winters, rainfall throughout the year with summer maximum. Characteristic of the warm temperate eastern margins of continents.

9. MEDITERRANEAN CLIMATE

Warm Temperate West Coast Climate - Köppen Code: Cs

Distribution

Occurs mainly between **30° and 45° latitudes on the western margins of continents**. Major regions: **Mediterranean Basin, California, Central Chile, Cape region of South Africa and south-western Australia**.

Cause of Formation

Develops due to the **seasonal shift of pressure and wind belts**. In summer, subtropical high-pressure belts shift poleward - descending air prevents rainfall. In winter, subtropical high shifts equatorward - westerlies and temperate cyclones bring rainfall.

Temperature and Rainfall

Parameter	Detail
Temperature	Summers warm to hot; winters mild. Small annual range due to maritime influence.
Annual Rainfall	Generally 400-900 mm.
Seasonality	Hot dry summer; mild wet winter - the most distinctive feature.
Nature of Rainfall	Winter rainfall is mainly cyclonic/frontal; also orographic on mountain slopes.

Vegetation, Soil and Agriculture

Natural vegetation: **sclerophyllous vegetation** - evergreen shrubs and open woodland. Known as **maquis** around the Mediterranean and **chaparral** in California. Plants have small, thick leathery leaves and deep roots. Characteristic soil: **terra rossa** (reddish soil over limestone). Agriculture: grapes, olives, citrus fruits and wheat. Irrigation widely used during dry summer.

Important Features

Hot dry summers and mild wet winters. Seasonal shift between subtropical high pressure and westerlies controls the rainfall pattern.

10. MARINE WEST COAST CLIMATE

British Type / Oceanic Climate - Köppen Code: Cfb

Distribution

Occurs mainly between **40° and 60° latitudes on the western margins of continents**. Major regions: British Isles, western France, north-western Europe, Pacific coast of north-western North America, southern Chile, Tasmania and New Zealand.

Cause of Formation

Develops because of the **strong maritime influence of oceans and the prevailing westerlies**. Westerlies carry moist oceanic air towards western continental margins throughout the year. Frequent temperate cyclones and fronts bring rainfall. Warm ocean currents further moderate temperatures, especially in western Europe.

Temperature and Rainfall

Parameter	Detail
Temperature	Summers cool to mild; winters mild. Small annual range due to oceanic influence.
Annual Rainfall	500-2,000 mm, occurring throughout the year. Many regions have a winter maximum.
Nature of Rainfall	Mainly frontal or cyclonic. Windward mountain slopes receive additional orographic rainfall.

Vegetation, Soil and Agriculture

Mainly **temperate deciduous forest** and mixed forest; also grasslands. Important trees: oak, beech, elm and maple. Mixed farming and **dairy farming** are important. Potatoes, oats, barley and fodder crops are widely cultivated.

Important Features

Mild summers, mild winters, small annual temperature range and rainfall throughout the year. Ocean, westerlies and temperate cyclones are its main climatic controls.

11. HUMID CONTINENTAL CLIMATE

Laurentian Type / Cool Temperate Eastern Margin - Köppen Code: Df

Distribution

Occurs mainly between **40° and 60°N on the eastern and interior parts of continents**. Almost absent in the Southern Hemisphere. Major regions: north-eastern USA, southern and south-eastern Canada, north-eastern China, Korea, northern Japan and parts of eastern Europe.

Cause of Formation

Develops because of **strong continental influence in the middle latitudes**. Land heats strongly in summer and cools rapidly in winter. The meeting of contrasting air masses (warm maritime vs cold continental/polar) produces frequent temperate cyclones and frontal rainfall.

Temperature and Rainfall

Parameter	Detail
Temperature	Summers are warm and short; winters long and cold. Temperatures frequently below freezing in winter.
Annual Temperature Range	Large because of strong continentality.
Annual Rainfall	Generally 500-1,000 mm, throughout the year. Summer maximum. Winter precipitation is often as snow.
Seasons	Four seasons clearly developed.

Vegetation, Soil and Agriculture

Mainly **mixed forests** of deciduous and coniferous trees. Important trees: oak, maple, beech, birch, pine and spruce. Crops: wheat, maize, soybean, potatoes and fodder crops. Dairy farming and livestock rearing are widespread.

Important Features

Warm summers, cold snowy winters, large annual temperature range and precipitation throughout the year. Strong continentality distinguishes it from the milder British Type climate.

12. SUBARCTIC CLIMATE

Taiga / Boreal Climate - Köppen Code: Dw

Distribution

Occurs mainly between **50° and 70°N**. Forms a broad belt across the northern parts of North America and Eurasia. Major regions: Siberia, northern Canada, Alaska and parts of Scandinavia.

Cause, Temperature and Rainfall

Develops in high-latitude continental interiors. Low solar angle and long winter nights cause severe cooling. Strong continentality further lowers winter temperatures. Winters are long and extremely cold; summers short and cool to mild. **Very large annual temperature range**, especially in the interiors of Siberia and Canada. Annual precipitation: **300-600 mm**, most during the short summer; winter precipitation as snow.

Vegetation, Soil and Agriculture

Characteristic vegetation: **Taiga or coniferous forest** - one of the largest continuous forest belts in the world. Trees: pine, spruce, fir and larch. Needle-shaped leaves reduce water loss; conical shape allows snow to slide off. Characteristic soil: **podzol** - acidic and strongly leached. Agriculture is limited by the short growing season. Lumbering, pulp and paper production, mining and fur-related activities are important.

Important Features

Long severe winters, short summers, low precipitation, very large annual temperature range and extensive coniferous forests.

13. TUNDRA CLIMATE

Köppen Code: ET

Distribution

Occurs mainly along the **Arctic margins** of North America, Europe and Asia: northern Alaska, northern Canada, coastal Greenland, northern Scandinavia and northern Siberia.

Cause, Temperature and Rainfall

At very high latitudes, the Sun remains low in the sky and solar radiation is weak. The warmest month remains between **0°C and 10°C**. Annual precipitation generally falls below **250 mm**; mainly as snow. Despite low precipitation, the surface can remain wet and marshy during summer because evaporation is very low and **permafrost** restricts drainage. (The ground below the surface remains permanently frozen - the active layer thaws only during summer.)

Vegetation, Soil and Key Features

Trees are generally absent. Vegetation consists mainly of mosses, lichens, sedges, grasses and dwarf shrubs. Plants remain short and close to the ground; shallow roots (permafrost prevents deep growth). Soil development is poor; waterlogging is common during summer.

Important Features

Very long cold winters, very short cool summer, low precipitation, permafrost and treeless vegetation.

14. POLAR ICE CAP CLIMATE

Köppen Code: EF

Distribution

Occurs mainly in **Antarctica and the interior of Greenland**. Small areas also occur on permanently ice-covered high Arctic islands.

Cause, Temperature and Rainfall

The highest latitudes receive weak solar radiation; long winter darkness; snow and ice have **high albedo** and reflect much incoming solar radiation. Temperatures remain **below 0°C throughout the year**. Antarctica records the **lowest temperatures on Earth**. Annual precipitation is very low, generally below 250 mm - these regions are considered **polar deserts**. Strong **katabatic winds** flow downslope from elevated interiors.

Key Features

Permanent snow and ice. Temperatures below freezing throughout the year. Extremely low precipitation. Virtually no vegetation. **Penguins** are characteristic of Antarctica; **polar bears** occur in the Arctic and not in Antarctica.

15. HIGHLAND CLIMATE

Mountain Climate - Köppen Code: H

Distribution

Occurs in high mountain regions irrespective of latitude. Major regions: **Himalayas, Andes, Rockies, Alps, Ethiopian Highlands and East African Highlands**.

Cause of Formation

Develops mainly because of **high altitude and relief**. Temperature decreases with increasing altitude at an average environmental lapse rate of about **6.5°C per 1,000 metres**. Mountains force moisture-bearing winds to rise, producing heavy orographic rainfall on windward slopes and dry conditions on leeward slopes (rain-shadow effect).

Altitudinal Zonation

Vegetation changes clearly with altitude - called **altitudinal zonation**. Lower slopes may support forests, followed by temperate or coniferous forests, alpine grasslands and finally permanent snow at the highest elevations.

Important Features

Controlled mainly by altitude and relief rather than latitude alone. Temperature decreases with height, rainfall differs between windward and leeward slopes, vegetation occurs in distinct vertical zones. Terrace farming is common in densely populated mountain regions.

LIMITATIONS OF KOPPEN'S CLIMATE CLASSIFICATION

Limitation	
Vegetation-based approach	Uses natural vegetation as an indicator of climate; does not directly explain the atmospheric processes responsible for a climate.
Sharp boundaries	Climate changes gradually from one region to another, but Koppen represents climatic regions using definite boundaries.
Temperature and rainfall dominate	Other climatic elements such as wind, humidity, cloudiness and sunshine receive relatively less importance.
Local climates are overlooked	Coastal influence, valleys, urban areas and local relief can produce climates different from the larger Koppen region.
Large climatic regions hide internal variations	Two places classified under the same Koppen type can still have noticeable differences in temperature and rainfall.

Koppen is excellent for showing the broad relationship between temperature, rainfall and vegetation, but is less effective in explaining climatic processes and local variations.

PREVIOUS YEAR QUESTIONS - PRELIMS

- Which one of the following is the characteristic climate of the Tropical Savanna Region? [2012]
 - Rainfall throughout the year
 - Rainfall in winter only
 - An extremely short dry season
 - A definite dry and wet season
- "The climate is extreme, rainfall is scanty and the people used to be nomadic herders." Which of the following regions is best described? [2013]
 - African Savanna
 - Central Asian Steppe
 - North American Prairie
 - Siberian Tundra
- The seasonal reversal of winds is the typical characteristic of: [2012]
 - Equatorial climate
 - Mediterranean climate
 - Monsoon climate
 - All of the above climates
- "Each day is more or less the same, the morning is clear and bright with a sea breeze; as the Sun climbs high in the sky, heat mounts up, dark clouds form, then rain comes with thunder and lightning. But the rain is soon over." Which of the following regions is described? [2015]
 - Savanna
 - Equatorial
 - Monsoon
 - Mediterranean
- Consider the following description:
Annual and daily range of temperatures is low. Precipitation occurs throughout the year. Precipitation varies between 50 cm-250 cm. What is this type of climate? [2024]
 - Equatorial climate
 - China type climate
 - Humid subtropical climate
 - Marine West coast climate

6. Assertion : Areas near the equator receive rainfall throughout the year.
Reason : High temperatures and high humidity cause convectional rain in most afternoons near the equator.
Which of the following statements is true ?
(a) Both A and R true; R is the correct explanation
(b) Both true; R is not a correct explanation
(c) Assertion is true, Reason is false
(d) Assertion is false, Reason is true
7. Consider the following statements:
1. In equatorial regions, the year is divided into four main seasons.
2. In the Mediterranean region, summer receives more rain.
3. In China type climates, rainfall occurs throughout the year.
4. Tropical highlands exhibit vertical zonation of different climates.
Which of the following statements is/are true ? [2002]
(a) 1, 2, 3 and 4
(b) 1, 2 and 3
(c) 1, 2 and 4
(d) 3 and 4
8. Assertion: Areas lying within 5-8 degrees latitude on either side of the equator receive rainfall throughout the year.
Reason: High temperatures and high humidity cause convectional rain to fall mostly in the afternoons near the equator.
Which of the following statements are correct ?
(a) Both A and R individually true; R is the correct explanation
(b) Both true; R is NOT the correct explanation
(c) A true, R false
(d) A false, R true
9. Consider the following statements:[2002]
1. In the world, the tropical deserts occur along the western margins of continents within the trade wind belt.
2. In India, the East Himalayan region gets high rainfall from north-east winds.
Which of the following statements are correct ?
(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

PREVIOUS YEAR QUESTIONS - MAINS

1. What characteristics can be assigned to the monsoon climate that succeeds in feeding more than 50 percent of the world population residing in Monsoon Asia? [2017]
2. Discuss the main features of the climatic classification system evolved by Koppen. Also point out its anomalies. [2005]
3. Write short notes on Cs type of climate, as per Koppen's classification. [1998]

NCERT Reference

Class 11th - Fundamentals of Physical Geography
Chapter 12: World Climate and Climate Change